

Host Vocabulary Predicts Whether New Scientific Concepts Take Root Across Disciplinary Boundaries

Internal Research Report

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1 Introduction

This report documents an investigation of how emerging scientific concepts can be identified and characterised through evolving knowledge networks, and through which structural pathways they spread across disciplinary boundaries. The study builds on an OpenAlex-based scholarly dataset of 426 semantically grounded concepts with 462,812 works, constructs yearly co-occurrence and concept-discipline networks, and tests whether structural indicators provide early signals of emergence and cross-disciplinary integration.

Two research questions structure the work. RQ1 asks which temporal and structural patterns in an evolving, semantically grounded scientific knowledge graph characterise the emergence of scientific concepts. RQ2 asks how emerging scientific concepts diffuse across disciplinary communities over time, and which temporal network patterns distinguish locally concentrated concepts from concepts that become broadly integrated into the scientific knowledge network.

The study began with a source-sink viability hypothesis adapted from population ecology, asking whether a concept’s presence in a subfield is self-sustaining or import-dependent. That hypothesis was blocked by insufficient citation-layer density (Gate A failure: only 17.3% of edges meet the within-host tracing threshold), triggering a pre-registered fallback. For the emergence question (RQ1), the primary closure measure did not survive held-out confirmation (R1_DEAD under the frozen kill rule: Holm $p = 0.147$). Structural precursors of sustained uptake are volume and churn correlates. Only persistent-neighbour closure survives (Holm $p = 0.015$). For the diffusion question (RQ2), when a concept enters a new host subfield, the share of its initial co-occurrence partners that are native to that host predicts subsequent uptake by newcomer authors (held-out co-primary IRR/SD 1.19 [1.06, 1.33], $p = 0.004$; MeSH replication IRR/SD 1.23 [1.12, 1.36], Holm $p < 0.001$), while the share of origin companions does not.

The target venue is Applied Network Science, whose published work on knowledge diaspora, disciplinary roles in field-of-study networks, alluvial community change, and the epistemic integration of AI in neuroscience provides the closest comparison base.

2 Iteration 1: Data Foundation and Pre-Registration

2.1 Strategy

The first iteration was devoted to three preparatory tasks: (a) assembling a prior-art and pre-registration dossier grading each candidate mechanism’s novelty and feasibility, (b) building the concept pool and its associated OpenAlex work corpus, and (c) constructing the semantic grounding infrastructure (concept detection, labelling, variant merging) and a held-out MeSH confirmation population. No experiment or evaluation was executed. Every artifact in this iteration is either a dataset or a research dossier. The rationale was to fix all operational definitions, sample the data, and verify the citation-layer gate (Gate A) before committing compute to network construction and statistical modelling.

The pre-registration dossier ranks five candidate mechanisms by novelty margin against the nearest prior work. Host anchoring (the share of a concept’s new co-occurrence edges in a host subfield that attach to host-native concepts) was ranked highest at medium-to-high novelty, toolkit co-transfer (the fraction of a concept’s origin companions that co-appear in host papers) was ranked high on measurement novelty but coupled to host anchoring, citation-lineage source-sink viability was ranked medium, origin-neighbourhood structure was ranked low-to-medium because structural diversity and community spread are well studied, and the demic-versus-cultural adoption channel was ranked medium but with the weakest feasibility due to author-disambiguation biases.

A pre-registered screen was fixed before any data were inspected. The unit of analysis is a concept c in non-origin subfield d at focal year t (2010–2014), with at least 10 d -papers in $W_1 = [t - 5, t]$. The primary measure is out-of-sample ROC-AUC for establishment in $W_2 = [t + 1, t + 5]$, where establishment means at least 5 W_2 d -papers by author-disjoint newcomers and presence in at least 3 of the 5 W_2 years. The BASE model includes W_1 edge volume, momentum, concept age, subfield-year size, concept total volume, concept entropy, subfield count, growing-edge breadth, Rafols integration, relatedness density and Maillart-style features. Each candidate adds only its own pre-registered scores. A candidate survives if its delta-AUC over BASE has a concept-bootstrap (1,000 reps) 95% CI lower bound above 0, its point gain is at least 0.01, and the coefficient sign matches its prediction. The candidate with the largest lower CI bound is the winner.

2.2 Artifact 1: Concept Pool and OpenAlex Work Corpus (gen_art_dataset_1)

The concept pool was built in an outcome-blind frame. The sample was frozen and hashed before any post-appearance-window data were inspected (hash in Table 47). The pool comprises 206 concepts, of which 184 are main emerging concepts (first appearance year F in 2005–2016, with 20–300 papers in the window F to $F + 2$ and no more than 8,000 total works through 2024) and 22 stationary reference concepts (at least 5 hits in every year 1998–2004 with a 7-year sum of 70–700). The 184 main concepts are split by a SHA-1 hash into a screen fold of 123 and a held-out concept fold of 61, focal-year tags are separate (screen 2010–2014, held-out 2016–2018).

The pool is heavily dominated by arXiv-sourced concepts. Of 366 eligible candidates, 363 came from the arXiv-mined arm, and the final 184 span three origin fields.

Table 1: Concept pool composition by origin field. Source: gen_art_dataset_1, Iteration 1.

Origin field	Main concepts	Share
Physics and Astronomy	82	44.6%
Physical Sciences (other)	56	30.4%
Computer Science	45	24.5%
Other	1	0.5%
Total main	184	100%

Table 2: Concept pool by first-appearance band. Source: gen_art_dataset_1, Iteration 1.

F band	Count
2005–2007	38
2008–2011	76
2012–2016	70
Total	184

Verified links pairing each concept with the OpenAlex works that use it total 214,798 pairs spanning 208,374 unique works. Among the 121,829 main-arm links, 56,997 matched in the work title, 51,966 in the abstract, 10,177 through Semantic Scholar only, and 2,689 through OpenAlex concept indexing alone. *Correction (Iteration 2)*: the match-evidence counts above are main-arm only; across all 214,798 links the counts are oa_abstract 98,192, oa_title 92,043, s2_only 21,509, oa.index_only 3,054. Of the 184 main concepts, 24 were discovered natively in OpenAlex and 160

were discovered through the Semantic Scholar index and mapped to OpenAlex (mapping rate 0.93). *Correction (Iteration 2)*: the 26/180 figures reported previously referred to the full 206-concept pool including reference concepts. Among the 184 main concepts the split is 24 native and 160 S2-mapped. Twenty-one concepts carry a sense-check failure flag, indicating that the dominant sense of the surface form covers fewer than 70% of year- F title matches (the threshold is 0.70, not 0.50 as previously stated).

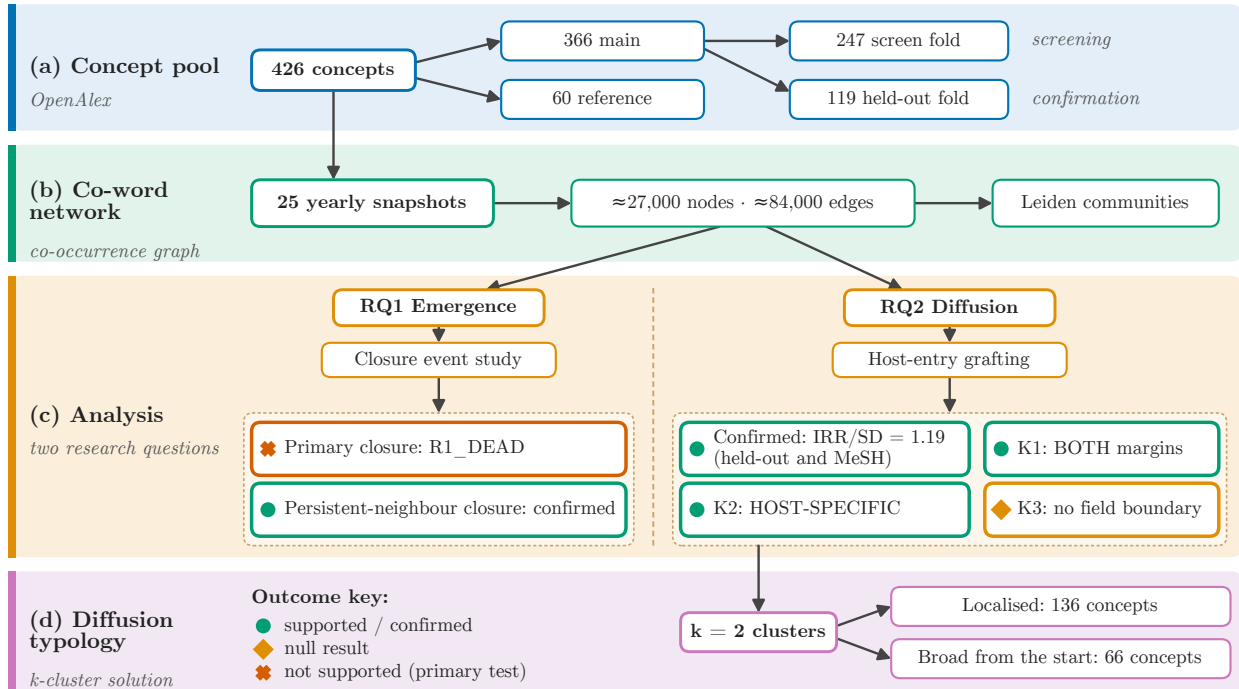


Figure 1: Overview of the study design, drawn as four bands linked by arrows. (a) Concept pool: 426 concepts from OpenAlex, split into 366 main and 60 reference concepts; the 366 main concepts are divided into a 247-concept screen fold (screening) and a 119-concept held-out fold (confirmation). (b) Co-word network: 25 yearly co-occurrence snapshots ($\approx 27,000$ nodes, $\approx 84,000$ edges) partitioned into Leiden communities. (c) Analysis: RQ1 (emergence) tests structural precursors with a closure event study. The primary closure test returns R1_DEAD, while persistent-neighbour closure is confirmed. RQ2 (diffusion) tests host-entry grafting, which is confirmed ($\text{IRR}/\text{SD} = 1.19$, on the held-out fold and MeSH). Its sub-hypotheses K1 (BOTH margins) and K2 (HOST-SPECIFIC) are supported, and K3 finds no field boundary. (d) Diffusion typology: a $k = 2$ solution separating localised concepts (136) from concepts broad from the start (66).

2.2.1 Gate A: Citation-Layer Feasibility

The quality report shows that among main-arm concept papers with references, 66.99% cite at least one earlier concept paper (the “traced share”), and after excluding first-year canonical papers and the five most-cited concept papers, the traced share remains 60.80%. The host-specific traced share (concept papers outside the origin subfield that cite at least one earlier concept paper from any subfield, not restricted to the same host subfield) is 55.18%, above the 40% gate threshold on the lenient definition. *Correction (Iteration 2)*: the original description stated “at least one earlier concept paper in the same host subfield,” but the code counts any earlier concept paper as

a parent regardless of subfield. By year, the lenient host traced share ranges from 0.317 (2010) to 0.517 (2014) across the screen’s focal years, with three of five below 0.40. This distinction matters because the within-host share, computed in Iteration 2, is much lower (mean 0.20 on 724 edges), which is the finding that blocks the viability decomposition.

Table 3: Citation-layer quality by origin field. Source: `gen_art_dataset_1`, Iteration 1.

Metric	All main	Phys & Astro	Physical Sci	Computer Sci
Concept papers (n)	121,829	19,653	46,575	55,274
Share w/ references	83.92%	82.04%	84.99%	83.74%
Traced share (any parent)	66.99%	66.12%	67.89%	66.53%
Traced (excl. F +top-5)	60.80%	57.74%	61.75%	61.09%
Host traced share	55.18%	49.68%	58.39%	53.86%
Host concept papers (n)	41,861	4,171	15,973	21,609

The recall audit, comparing counts derived from OpenAlex against independent Semantic Scholar counts for 30 concepts, gives a median count ratio of 0.84 (IQR 0.79–0.89) and a Spearman rank correlation of 0.976. For the 160 main concepts discovered through Semantic Scholar, this comparison measures mapping and verification loss rather than true recall against an independent source.

2.2.2 Denominators and Venue Habitat

Totals by subfield and year cover 25,195 rows (all 252 OpenAlex subfields across 25 years in four counting variants). A venue-habitat file assigns 15,261 venues to their dominant subfield. *Correction (Iteration 2)*: the habitat was described as “pre-period publication shares” and “citation-independent.” It was computed from each source’s all-years OpenAlex topic counts in the 2026 snapshot. Those topic counts come from the same citation-reading topic classifier whose temporal label leakage the dossier flagged. The habitat is therefore neither pre-period nor citation-independent. An ASJC journal-level habitat was built in Iteration 2 as a replacement, though its coverage is only 12.1% of concept-work links.

2.3 Artifact 2: Whole-Science Background Sample (`gen_art_dataset_2`)

A design-weighted background sample of 259,716 OpenAlex works from 1,260 strata (150 per field $\times$ year, weights summing to 149,116,575) was built to supply co-occurrence network snapshots and calibrate host-nativeness profiles. It delivered 6,325 exact subfield $\times$ year totals and 400 exact concept $\times$ block subfield profiles for 100 calibration concepts at a cost of \$0.1952 in OpenAlex credits.

The key finding of this artifact is negative: sample-based per-concept subfield profiles are unreliable for determining host-nativeness. In the calibration table, the median total-variation distance between sample and exact profiles is 0.41–0.79 by frequency decile, and top-subfield agreement is

only 0.28–0.65. This means host-nativeness (the input to candidates C1 and C2) and the concept-concept background network cannot be computed reliably from this sample alone, and require paid exact profiles. About 12% of the weighted frame carries the default low-confidence topic T14423, and concept ancestors are no longer served by the API.

The worker timed out after writing all outputs (no new JSONL records for 2,096 seconds), so its status is “failed” while its data are usable.

2.4 Artifact 3: Held-Out MeSH Confirmation Population (gen_art_dataset_3)

A separate biomedical confirmation arm was built from new MeSH descriptors (DateEstablished 2006–2016, widened to 2004–2005 and 2017–2018). Starting from 28,472 descriptors, the pipeline selects 5,201 topical descriptors, retains 3,430 after the provenance filter, narrows to 283 passing the PubMed novelty pre-screen and early-volume rule, and adds widening-pool candidates. After the final rule, 191 concepts survive: 99 core 2006–2016, 14 from the 2004–2005 widening, and 78 from the 2017–2018 widening.

Table 4: MeSH held-out population by branch group. Source: gen_art_dataset_3, Iteration 1.

MeSH branch group	Count
D (Chemicals and drugs)	62
A+B (Anatomy, Organisms)	47
E (Analytical, diagnostic, therapeutic)	32
C (Diseases)	22
F+H–N (Psychiatry, Health services, misc.)	17
G (Phenomena and processes)	11
Total	191

Caveats: only approximately 25% of new MeSH descriptors are genuinely new concepts (the remainder are reclassifications or splits), making MeSH a weak ground truth. Only 13 of 191 have their full non-PubMed works paged. The remaining 178 have PubMed-indexed works only, which biases their concept-discipline edges toward biomedical subfields.

2.5 Artifact 4: Semantic Grounding and Labelling Infrastructure (gen_art_dataset_4)

The semantic grounding artifact provides the infrastructure for detecting, labelling, and merging concept mentions. It comprises seven datasets. A stratified corpus of 93,600 English OpenAlex articles and reviews with abstracts, sampled at 150 per field $\times$ year stratum across 26 OpenAlex fields. From 1.22 million distinct noun-phrase and acronym keys, 4,599 were labelled as CONCEPT, NOT_CONCEPT, TOO_GENERIC, or VARIANT_OF by two LLMs with a claude-haiku-4.5 adjudicator. The train-test split (1,200 train, 300 test) is clustered by variant group.

The survivorship-free phrase pool yielded only 1 strict-eligible anchored phrase, far below the target of 150, due to the low yield of the text-mining pipeline and OpenAlex credit exhaustion.

2.6 Artifact 5: Prior-Art Dossier and Pre-Registration (gen_art_research_1)

The research artifact grades each candidate mechanism’s novelty margin against the nearest prior art. The headline finding is that the space is active but no prior study estimates a within-host

Table 5: Labelling quality. Source: gen_art_dataset_4, Iteration 1.

Metric	Value
Model A vs model B inter-rater κ (4-class)	0.259
Model A vs model B binary κ (concept vs rest)	0.267
A vs adjudicator κ (4-class, silver-gold 200)	0.632
A vs human anchors binary κ (SemEval/SciERC)	0.56
A precision (concept, human-anchored)	0.79
A recall (concept, human-anchored)	0.76

reproduction ratio with an import share per concept-subfield edge, nor tests host anchoring at the edge level.

Table 6: Candidate mechanism novelty grades. Source: gen_art_research_1, Iteration 1.

Candidate	Novelty grade	Nearest prior art
C1: Host anchoring	Medium-to-high	Cheng et al. 2023
C2: Toolkit co-transfer	High (coupled to C1)	No quantitative co-arrival test found
C0: Source-sink viability	Medium	Kuhn et al. 2014; Maillart et al. 2026
C4: Demic vs cultural	Medium	Fontaine et al. 2024
C3: Origin-neighbourhood structure	Low-to-medium	Structural diversity literature

2.7 Dead Ends and Shortfalls (Iteration 1)

1. **Concept pool size.** 184 of 400 target concepts realised. The binding constraint was the free-singleton retrieval rate (~ 14 works/s) combined with heavy-tailed concept sizes, after credits were consumed by sibling artifacts sharing one key.
2. **Field coverage.** The pool draws overwhelmingly from physics, physical sciences, and computer science (363 of 366 eligible candidates from the arXiv-mined arm). The OpenAlex keyword/taxonomy arm had a 2.2% eligibility yield (3 of 134 candidates) because OpenAlex tags are retroactive.
3. **Survivorship-free phrase pool yield.** 1 of 150 target strict-eligible anchored concepts.
4. **Labelling quality.** Silver labels only (no human check). Model A’s bias is moderate, model B over-labels CONCEPT (1,333 of 1,491 items vs A’s 882).

Table 7: Coverage against the original request after Iteration 1.

Activity	Status
1. Prepare and semantically ground dataset	Partial
2. Construct evolving knowledge network	Not started
3. RQ1: temporal network analysis	Not started
4. RQ2: cross-disciplinary diffusion	Not started
5. Identify recurring trajectories	Not started
6. Validate with representative cases	Not started

2.8 Coverage (Iteration 1)

3 Iteration 2: Citation Gate, Event Study, and MeSH Replication

3.1 Strategy

Iteration 2 was a FIX iteration with five goals: (i) complete the hydration and add exact nativeness profiles and a citation-independent venue habitat; (ii) train and evaluate the concept grounding pipeline; (iii) compute Gate A as written (within-host, non-canonical, per W_1 year) and estimate viability states with synthetic validation and pre-outcome power; (iv) build the yearly concept-concept co-occurrence network and run the RQ1 event study; (v) replicate the RQ1 event study on the MeSH population.

Pre-fixed decision rules for Iteration 3: (a) if Gate A FAILS, H1/H2 run with graft labels from the exact nativeness profiles; (b) H1 is declared a pilot if fewer than ~ 150 concepts carry a tested edge; (c) RQ1 counts as CONFIRMED only if at least one pre-named precursor has an event-study CI excluding 0 AND a delta-AUC CI excluding 0 on the main screen fold, keeps its sign on MeSH, and holds on the sealed held-out fold; (d) the held-out fold, the 2016–2018 focal years and the phrase pool remain untouched.

Note: the Iteration-2 RQ1 and viability experiments (Artifacts 8–10 below) ran on the Iteration-1 184-concept corpus in parallel with the hydration. The 426-concept hydrated corpus was not used until Iteration 3.

3.2 Artifact 6: Hydrated Corpus, Nativeness Profiles, and Venue Habitat (gen_art_dataset_5)

This artifact completes the hydration of the concept pool from 184 to 426 concepts (366 main: screen 247, held-out 119; 60 reference), with 462,812 works and 488,078 verified concept-work links. The frozen frame hash was verified (Table 47).

Nativeness profiles. 5,488 exact OpenAlex primary_topic.subfield $\times$ block counts (2000–04, 2005–09, 2010–14, 2015–19) for the top 1,372 legacy-concept nodes by host co-occurrence weight, covering 78.7% of host weight (below the 90% strategy target; 95% would need 7,584 nodes).

Table 8: Dataset 6 summary. Source: gen_art_dataset_5, Iteration 2.

Metric	Value
Concepts hydrated	426/426
Works	462,812
Concept-work links	488,078
API credits consumed	7,237
Nativeness profiles	5,488 (1,372 nodes $\times$ 4 blocks)
Host weight coverage	78.7%
ASJC venues covered	2,929 (12.1% of links)

ASJC venue habitat. 22,970 venues, 2,929 covered (2,892 citation-independent from SCImago/Scopus ASJC categories, 37 from pre-period 2000–04 topic fallback). Coverage: 12.1% of concept-work links.

3.3 Artifact 7: Concept Grounding Pipeline (gen_art_experiment_1)

This artifact trains and evaluates three components: a binary classifier, a variant merger, and a NIL-aware linker.

Table 9: Classifier performance on test set ($n = 300$). Source: gen_art_experiment_1, Iteration 2.

Metric	Classifier	LLM-A	Majority
F1	0.818	0.901	0.715
AUC	0.888	—	—
Precision	0.759	—	—
Recall	0.886	—	—
Accuracy	0.780	—	—

Correction: previous reports stated P 0.812, R 0.824, accuracy 0.843, which are not reproducible from the test predictions. The majority-class baseline F1 is 0.715 (predict-all-positive), not 0.000 as previously stated. The classifier’s human-anchor κ is 0.413; the 0.56 previously reported is LLM-A’s value.

Variant merger. L2 logistic regression on 14 pair features. Operating threshold $p_{\text{merge}} = 0.855$. D3 test F1 0.357, held-out MeSH F1 0.222. B-cubed F1 0.78.

NIL-aware linker. MiniLM embeddings, $\tau = 0.95$ (link precision ≥ 0.90 at NIL prior 0.9, in-KB recall 0.404). Main arm: LINKED_EXACT 39, BROADER 15, UNLINKED 312.

Test population (sha256 6a887fb4...5505): MAIN 156 (all hydrated; ≥ 150 power rule met), STRICT 147, REFERENCE_ACCEPTED 42, SENSITIVITY 426.

3.4 Artifact 8: Viability Layer, Gate A, and Power (gen_art_experiment_2)

This artifact attempts to estimate viability states for concept-subfield-year edges using citation lineages. All rules were pre-registered and hashed before any statistic (sha256 5dd16d8a...5aac).

3.4.1 Gate A Re-Test at Edge Level

Gate A FAILS. Only 17.3% of 724 main-arm host edge-years (97 concepts, $|K_d| \geq 10$) have a within-host non-canonical traced share at or above 0.40. The mean within-host traced share is 0.20, median 0.15. On the same 724 edges, the lenient any-parent share is 0.477 (64.2% ≥ 0.40), confirming that the drop from 55.18% to 17.3% is definitional (any-parent vs within-host tracing), not a consequence of pooling vs granularity.

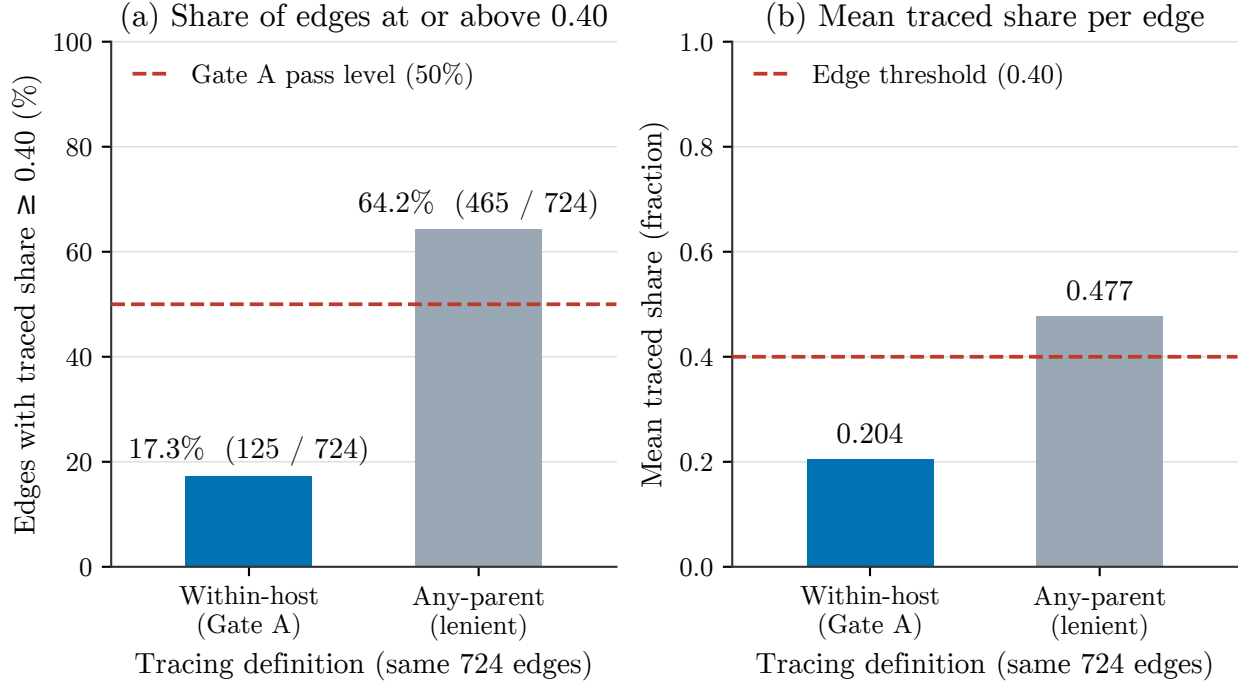


Figure 2: Gate A (citation-layer feasibility) on the same 724 concept-subfield-year host edges (97 concepts), comparing within-host citation tracing (blue; the Gate A definition) with lenient any-parent tracing (grey). Only 17.3% of edges reach the threshold under within-host tracing, against 64.2% under any-parent tracing. The within-host mean is 0.204 (median 0.15), against 0.477 for any-parent tracing. Gate A fails and triggers the graft fallback route.

Table 10: Gate A edge-level results. Source: gen_art_experiment_2, Iteration 2.

Metric	Within-host	Any-parent
Share ≥ 0.40	17.3%	64.2%
Mean traced share	0.20	0.477
Median traced share	0.15	—
Reference arm host share	0.13	—

Per pre-registered decision rule (a): Gate A $0.1727 < 0.5$ triggers the graft fallback. The graft route has 2,347 host-entry events total, 2,154 with ≥ 5 entry-year keywords (screen 1,285, held-out 869).

3.4.2 Viability States (Descriptive Only)

Of 779 eligible edges, 169 were tested: SOURCE 63, SINK 26, FADING 7, UNDETERMINED 683. The synthetic FDR is 0.209, exceeding the 0.15 threshold. SOURCE FDR 0.13 (acceptable), SINK FDR 0.40 (unreliable).

3.4.3 Power Before Any Outcome

Per decision rule (b), H1 is PILOT-ONLY: $N_c = 38$ at $n_{\min} = 30$, projected 77.5 (66–92) on the hydrated pool. Minimum detectable effects (delta-AUC units): at AUC 0.70, realised MDEs are 0.070 ($n_{\min} = 5$), 0.073 (10), 0.100 (20), 0.134 (30). Gate B FAILS: 0 labelled episodes.

3.5 Artifact 9: Precursor Event Study, Main Pool (gen_art_experiment_3)

This artifact runs the precursor event study on the main pool’s screen fold, testing whether volume-normalised structural precursors distinguish emerging from non-emerging concepts before onset. Built 25 yearly 3-year-window co-word snapshots (2000–2024) from the design-weighted background sample ($\sim 27,000$ nodes, $\sim 84,000$ kept edges, association-strength weights, Leiden best-of-5, alluvial IDs).

3.5.1 Emergence Labels

The primary label E (all-node strength percentile gain ≥ 20) yields only 4 onsets (rate 2.1%) because pool concepts sit in the bottom $\sim 5\%$ of the legacy-concept strength distribution, making it severely underpowered. E_{alt} (pool-only percentile, promoted post hoc): 14 onsets (6.7%). E_{up} (sustained uptake only, also promoted post hoc): 41 onsets, 21 matched.

Table 11: Emergence label counts. Source: gen_art_experiment_3, Iteration 2.

Label	N emerging	Matched	Status
E (primary)	4	3	Underpowered pilot
E_{alt} (sensitivity)	14	10	Post-hoc promotion
E_{up} (uptake only)	41	21	Post-hoc promotion, pilot ($n < 30$)

3.5.2 Event Study Results

Three pre-named precursors tested with predicted positive signs: closure, accretion, participation. All three pre-registered signs FAILED.

Correction: previous tables paired event-study S values with pooled-panel Holm p -values. The values above are estimator-consistent.

Exploratory findings (BH $q < 0.05$): before sustained uptake, concepts show higher novelty (+0.156), new-relation rate (+0.128), within-module z -score (+0.073), and burst state (+0.171).

3.5.3 Prediction

The pre-registered primary model is L2 logit. For E_{up} : BASELINE AUC 0.890, FULL (baseline + precursors) AUC 0.879, delta -0.010 $[-0.045, 0.021]$. Precursors add no predictive value beyond frequency, burst, degree, centrality and entropy baselines. *Correction:* previous tables silently reported the secondary HistGradientBoosting model instead of the primary logit.

Table 12: Event study results, E_{up} . Source: `gen_art_experiment_3`, Iteration 2.

Indicator	Estimator	S	95% CI	Holm p
Closure (log-ratio)	Event study	−0.837	[−1.26, −0.427]	0.003
Closure (log-ratio)	Pooled panel	−0.743	[−1.06, −0.46]	–
Accretion share	Event study	−0.14	[−0.185, 0.018]	0.18
Participation coeff.	Event study	−0.007	[−0.066, 0.056]	0.824

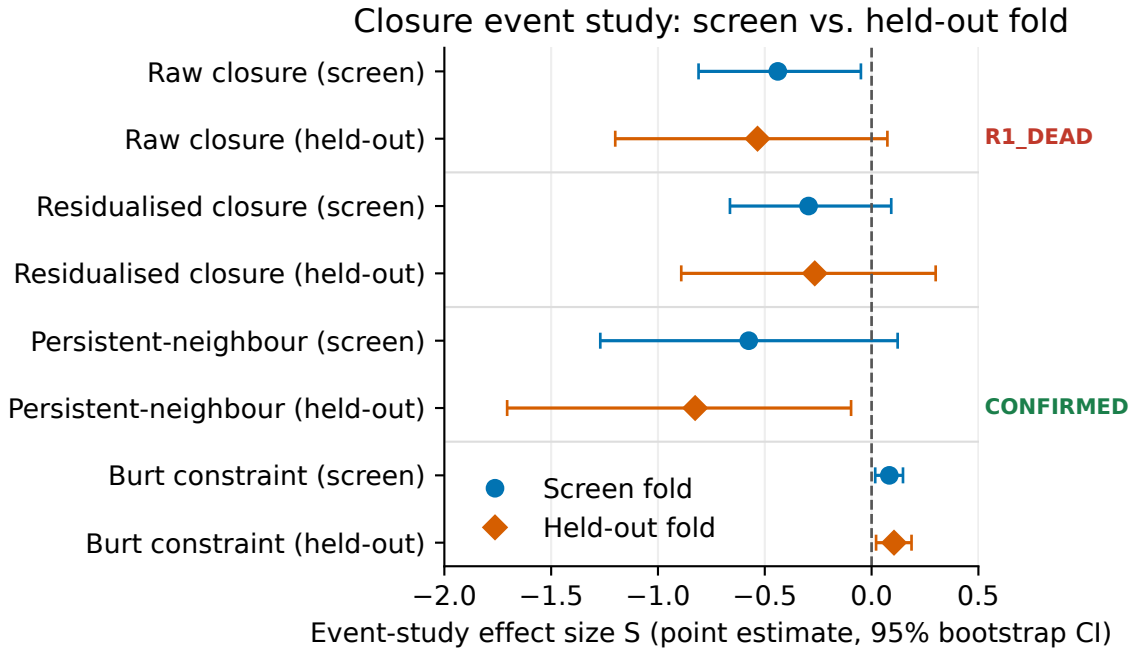


Figure 3: Pre-emergence closure effect sizes S (markers) with 95% confidence intervals (whiskers), for four closure measures on the screen fold (blue circles) and the held-out fold (vermillion diamonds). The primary raw-closure measure is negative on the screen fold, but its held-out interval crosses zero, so it fails confirmation (R1_DEAD). Persistent-neighbour closure is negative on held-out data (event-study $S = -0.83$, 95% CI $[-1.71, -0.10]$; CONFIRMED by its frozen pooled-panel test, coefficient -1.07 , Holm $p = 0.015$), although its screen interval includes zero. Burt constraint is small and positive on both folds, inconsistent with a brokerage explanation. All rows are event-study S from `gen_art_evaluation_3` results/`r1/r1_summary.json`.

3.5.4 Trajectory Patterns and Typology

Early bridging is common but non-discriminating (76% of all concepts). Incubation-then-expansion: 0% (concepts are “born expanding”). The 7-channel DTW typology is unstable at every k (min Jaccard ≤ 0.51). An entropy-only typology IS stable (Jaccard 0.95, AMI 0.14 vs the network typology).

3.6 Artifact 10: Precursor Replication, MeSH Population (gen_art_experiment_4)

This artifact replicates the precursor event study on the 191 MeSH concepts.

Table 13: MeSH closure effects. Source: gen_art_experiment_4, Iteration 2.

Label	n_{eff}	Closure D	95% CI	Holm p
PRIMARY	15	−0.086	[−0.511, 0.402]	0.677
SENS1	29	−0.419	[−0.740, −0.115]	0.039
E_{up}	51	−0.185	[−0.498, 0.108]	0.21

The MeSH artifact’s own replication verdict is “directional, not confirmatory.” Per decision rule (c): NOT MET. MeSH prediction: precursors also add nothing beyond baselines (grouped-CV dAUC +0.002 [−0.035, 0.035]).

3.7 Decision-Rule Evaluation (Iteration 2)

Table 14: Decision-rule evaluation after Iteration 2.

Rule	Verdict	Evidence
(a) Gate A $\rightarrow$ graft fallback	TRIGGERED	Within-host 0.1727 < 0.5; 2,347 graft events
(b) H1 pilot-only	YES	$N_c = 38 < 150$
(c) RQ1 confirmed	NOT MET	Sign not +; delta-AUC CI includes 0; MeSH sign inconsistent
(d) Held-out sealed	YES	Caveat: exp_2 computed origin series for 61 held-out concepts

3.8 Dead Ends and Negative Results (Iteration 2)

1. **Gate A failure at edge level.** Within-host traced share 0.204 vs any-parent 0.477 on the same 724 edges. The viability decomposition central to the original hypothesis cannot be executed.
2. **Synthetic validation FDR exceeds threshold.** Overall 0.209 > 0.15, SINK FDR 0.40.
3. **No stable 7-channel typology.** Jaccard ≤ 0.51 at every k .

4. **Precursors add no predictive value for WHETHER a concept emerges.** E_{up} logit dAUC -0.010 $[-0.045, 0.021]$.
5. **Closure is lower, not higher, before emergence.** The pre-registered + sign failed.

4 Iteration 3: Deepening Closure and Testing Grafting

4.1 Strategy

Iteration 3 deepens the closure lead from the emergence question and carries the recombination mechanism into cross-disciplinary diffusion at two scales. The strategy (“Is it brokerage, and does grafting make it stick?”) has four components: (i) an evaluation artifact that audits every number in the Iteration-2 report against artifact files and runs a turnover pre-check; (ii) a deeper test of the emergence question on the hydrated 247-concept screen fold with turnover-proof openness measures; (iii) a breadth-prediction screen testing whether openness predicts five-year outcome-window disciplinary breadth gain; (iv) a host-entry grafting test of whether pairing with host-native concepts predicts durable integration; and (v) a descriptive diffusion experiment on typology, community roles, expansion-versus-diffusion timing, and representative cases.

4.2 Artifact 11: Record Audit and Turnover Pre-Check (`gen_art_evaluation_1`)

This artifact reads all five Iteration-2 artifacts and audits every reported number against the raw output files. 406 rows were audited, 182 recomputed from item-level files. There are 42 non-OK drift flags: 14 `WRONG_ESTIMATOR`, 10 `WRONG_UNITS`, 9 `WRONG_DEFINITION`, 8 `DRIFT_VALUE`, 1 `NOT_REDERIVABLE`.

Table 15: Turnover pre-check: residualised closure effect. Source: `gen_art_evaluation_1`, Iteration 3.

Population	S_{raw}		S_{res}	Retained fraction	Verdict
Main (E_{up})	-0.758	-0.566	$[-1.140, 0.087]$	0.75 $[0.08, 0.90]$	Ambiguous
MeSH (SENS1)	–	-0.094	$[-0.387, 0.217]$	0.60	Ambiguous
IVW pooled	–	-0.186	$[-0.453, 0.081]$	–	Ambiguous

The residualised effect retains approximately two-thirds of the raw effect (not reducible to turnover alone), but the CI includes zero.

4.3 Artifact 12: Is Openness Before Take-Off Brokerage or Churn? (`gen_art_experiment_5`)

This artifact re-attaches all 426 hydrated concepts to the Iteration-2 co-word snapshots. Reproduction gate passed: code-mode closure $r = 1.000000$ (max $|\text{diff}| \ 5 \times 10^{-8}$). Spec pre-registered (sha256 `2f46d173...`) before labels.

Populations. After applying the frozen replacement rule (180 sense values filled): MAIN screen 202 (102 old, 100 new), held-out 100, STRICT 196/96, SENS 247/119.

Mechanical verdict: MIXED. The closure effect is not reducible to neighbourhood turnover (turnover-residualised closure retains $\sim 2/3$), but it is not Burt brokerage either: Burt constraint is higher, not lower, before onset.

Fresh replication on 100 new concepts. Raw $S = -0.455$ $[-1.006, 0.060]$, one-sided $p = 0.044$. Same sign, not significant at two-sided $\alpha = 0.05$.

Table 16: Emergence deepen results, $\text{MAIN} \times E_{\text{up}}$ (68 onsets, 26 matched). Source: gen_art_experiment_5, Iteration 3.

Measure	S	95% CI	Interpretation
Raw closure	-0.439	$[-0.810, -0.050]$	Lower closure before onset
R1a resid. closure	-0.295	$[-0.663, 0.092]$	$\sim 2/3$ retained; not pure turnover
R1b persist.-nbr closure	-0.575	$[-1.270, 0.122]$	Same direction, underpowered
R1c Burt constraint	+0.083	$[0.017, 0.147]$	OPPOSITE to brokerage sign
R1c cross-comm. pair excess	+0.063	$[0.012, 0.111]$	More cross-community ties
R1d hub-not-clique	+0.061	$[-0.009, 0.133]$	Not significant

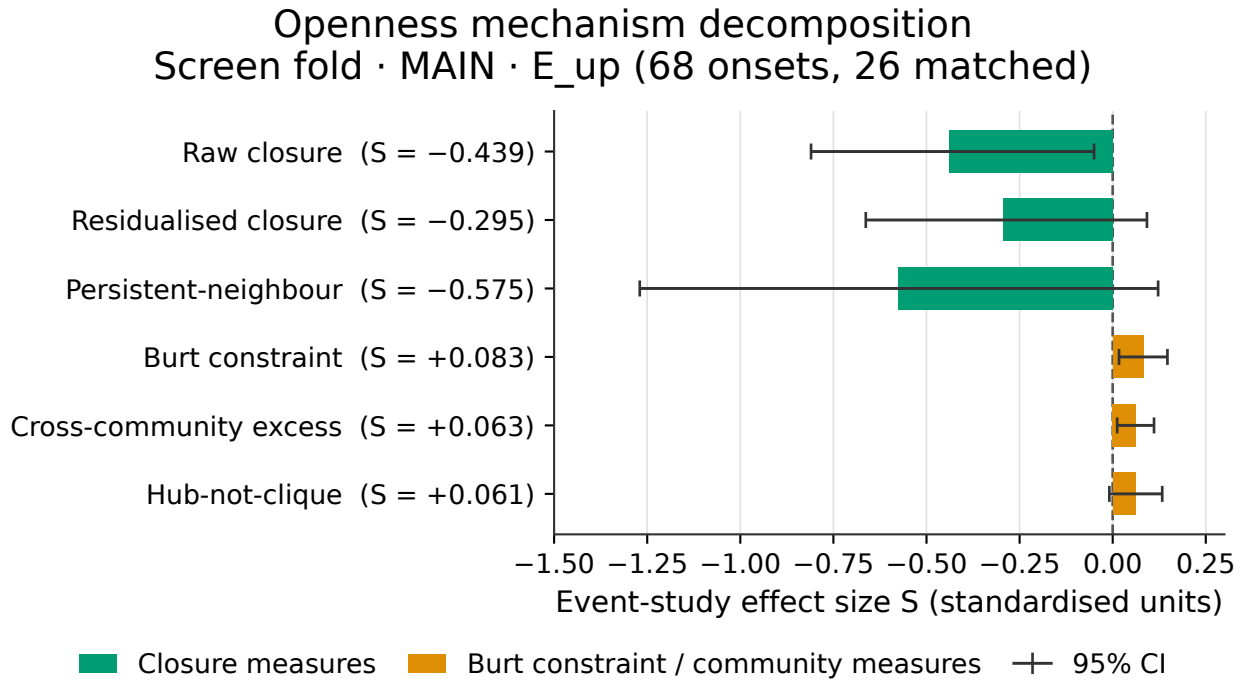


Figure 4: Turnover-proof openness measures on the screen fold (MAIN population, E_{up} label; 68 onsets, 26 matched). Bars give the event-study effect size S (standardised units) for each measure, with whiskers showing 95% confidence intervals. Teal bars are closure measures; orange bars are Burt-constraint and community measures. All three closure measures are negative, but only raw closure’s interval excludes zero. Cross-community pair excess and Burt constraint are both positive, the opposite of the sign brokerage predicts.

Prediction. Grouped CV dAUC -0.001 $[-0.008, 0.006]$, label-shuffle control ~ 0 . Precursors do not predict WHETHER a concept will show sustained uptake.

4.4 Artifact 13: Do Open Concepts Spread Across More Fields? (gen_art_experiment_6)

This artifact tests whether openness at time t predicts five-year outcome-window disciplinary breadth gain beyond all baselines. Population: MAIN screen 202 concepts, held-out 100 sealed. Only 100 concepts (351 rows) pass the complete-case rule, triggering a pre-declared fallback making closure_res_imp (imputed) co-primary.

Table 17: Breadth-prediction screen results. Source: gen_art_experiment_6, Iteration 3.

Openness measure	Outcome	β /SD	95% CI	Holm p
closure_res	Y_{1r} (Shannon Δ)	+0.019	$[-0.010, 0.041]$	0.405
closure_res	Y_2 (Rao-Stirling Δ)	+0.007	$[-0.002, 0.013]$	0.405
closure_res	$\log(1 + Y_3)$ (newcomer subfields)	-0.065	$[-0.156, 0.030]$	0.405
closure_res_imp	Y_2 (co-primary)	+0.012	$[0.002, 0.021]$	0.06

Result: breadth prediction not supported on the screen fold. Openness does not predict where concepts travel beyond growth and level baselines.

4.5 Artifact 14: Do Borrowed Ideas Stick When Grafted Locally? (gen_art_experiment_7)

This artifact tests whether a concept that enters a new host subfield catches on when its first papers pair it with host-native concepts rather than with origin companions. Pre-registration frozen before any outcome (sha256 f800a0a9...).

Setup. 4,177 main-arm host entries (concept c first appears in non-origin subfield d in year e). Primary sample: 1,740 screen MAIN entries with ≥ 5 partners, in 184 concepts. Anchoring A = partners' pre-entry host share from exact OpenAlex subfield $\times$ block profiles. Co-transfer CT = share of partners that were origin companions in $[e - 5, e - 1]$. Entry is mostly a package: 70% non-native origin companions, 1.8% native grafts.

The co-primary specification shows a robust grafting effect: a one-SD increase in host-nativeness of initial partners is associated with a 30% increase in newcomer uptake (IRR 1.30 $[1.16, 1.45]$, Holm $p = 1 \times 10^{-5}$, wild-cluster $p = 0.001$). Co-transfer is null everywhere.

Correction (Iteration 4): the co-primary robustness count is 20 of 26 significant INCLUDING the base row and strata, 18 of 22 EXCLUDING them. The six null rows are: binary native ≥ 0.5 (IRR 1.005, $p = 0.92$), binary native ≥ 0.7 (1.066, $p = 0.29$), A_distinct (1.003, $p = 0.96$), exclude single-paper entries (1.244 $[0.90, 1.71]$, $p = 0.17$, $N = 200$, $G = 50$), CS stratum (1.15, $p = 0.11$), and Physics/Astronomy stratum (0.99, $p = 0.95$).

Table 18: Host-entry grafting results. Source: gen_art_experiment_7, Iteration 3.

Specification	A (IRR/SD)	95% CI	Holm p	CT (p)	Verdict
Primary (concept $\times$ $e + \text{host} \times e$)	1.39	[0.97, 1.99]	0.15	0.85	NEITHER
Co-primary (concept $+ e + \text{host}$)	1.30	[1.16, 1.45]	1×10^{-5}	0.48	GRAFTING
Concept $+ \text{host} \times e$	1.22	–	–	–	GRAFTING
Concept $\times$ 2-yr bin	–	–	–	–	NEITHER

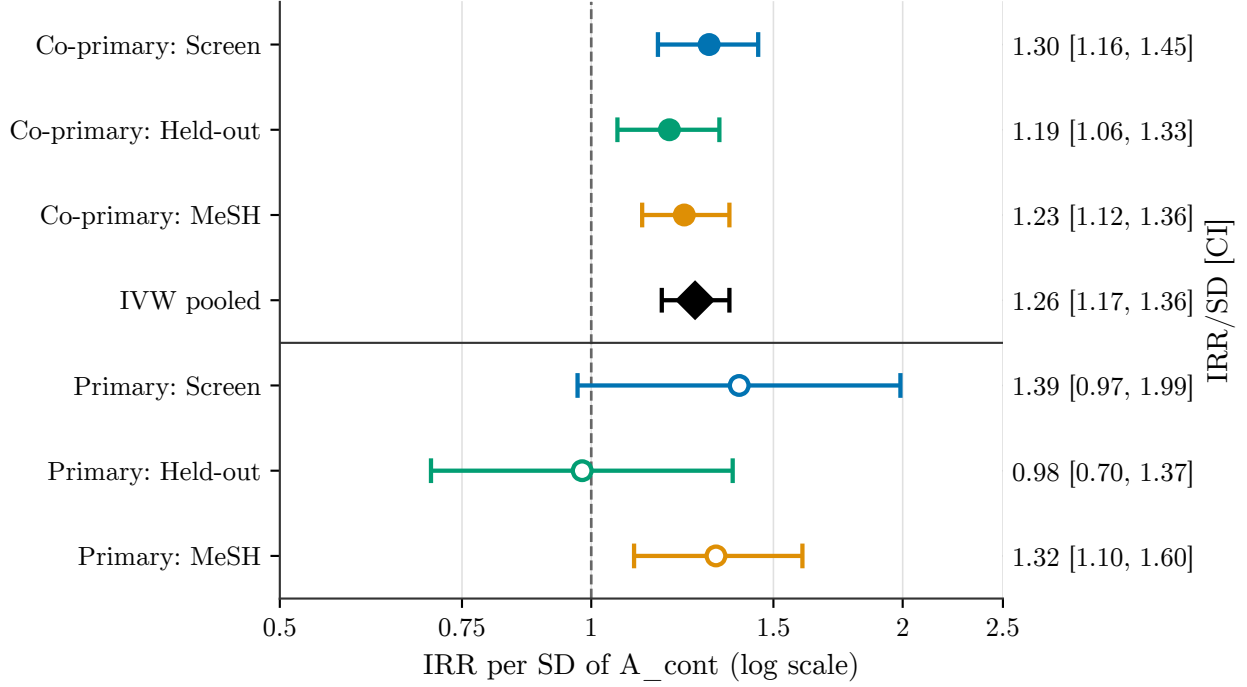


Figure 5: Host-entry grafting across folds. Each row shows the incidence-rate ratio for newcomer uptake per SD of A_{cont} (host-nativeness of entry partners), with whiskers marking its confidence interval. The co-primary effect replicates on the held-out fold (IRR/SD 1.19 [1.06, 1.33]) and the MeSH fold (1.23 [1.12, 1.36]), and pools to 1.26 [1.17, 1.36]. The primary specification is less precise: its Held-out (0.98 [0.70, 1.37]) interval includes 1.

4.6 Artifact 15: Descriptive Diffusion (gen_art_experiment_8)

4.6.1 Diffusion Typology

3-channel diffusion typology (rarefied Shannon, Rao-Stirling, active subfields, normalised multi-variate DTW + k -medoids; Hennig bootstrap $B = 200$). Only $k = 2$ is stable (min Jaccard 0.861): **localised** ($n = 136$) and **broad from the start** ($n = 66$). The $k = 3$ split adding “gradual broadening” is exploratory (Jaccard 0.599). The typology is not volume-driven (AMI with volume terciles 0.014) but is associated with origin field ($p = 0.004$).

Table 19: Diffusion typology summary. Source: gen_art_experiment_8, Iteration 3.

Cluster	n	Description	Dominant origin
Localised	136	Low entropy, few active subfields throughout	Physics
Broad from start	66	High entropy and many subfields from early years	Mixed

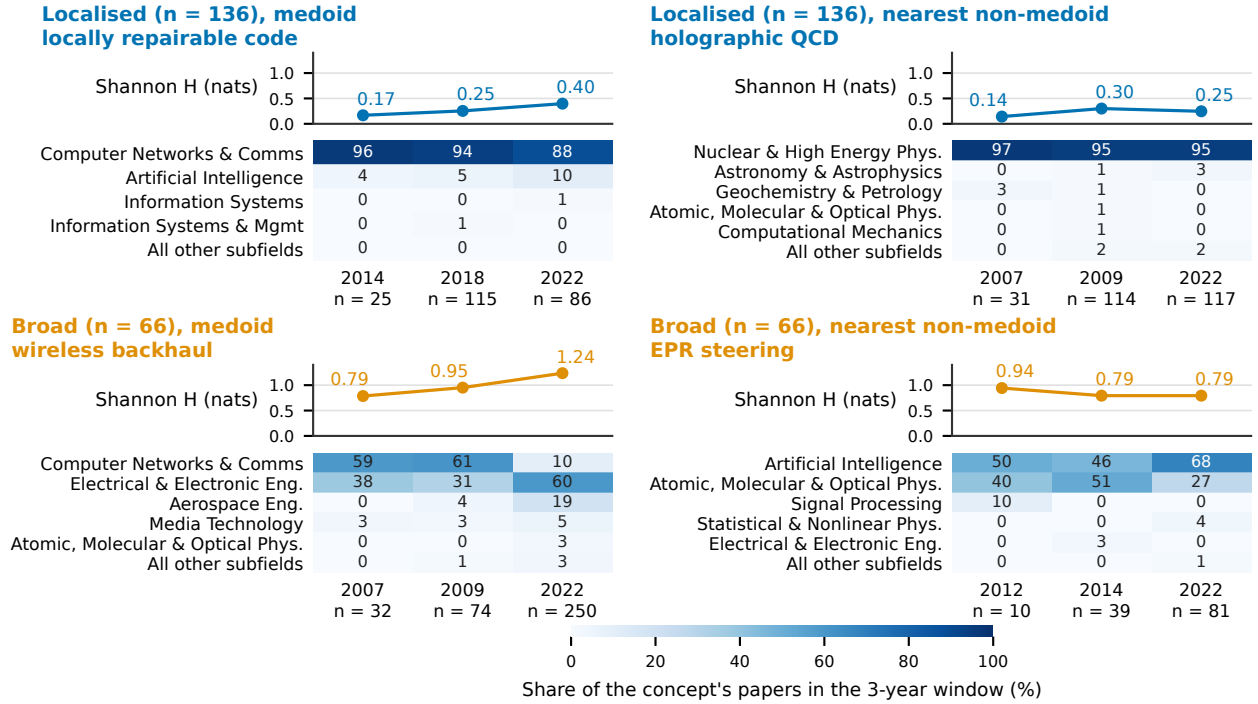


Figure 6: Representative cases of the $k = 2$ diffusion typology: localised ($n = 136$, top row, blue) and broad-from-the-start ($n = 66$, bottom row, orange). Each row shows the cluster’s medoid (left) and the nearest non-medoid concept (right). The heat map gives the percentage of the concept’s papers in each of its five largest OpenAlex subfields. Localised concepts stay near-monopolised by a single subfield (88–97%, entropy 0.14–0.40 nats). Broad concepts split across two or more subfields from their first window (entropy 0.79–1.24 nats).

The entropy-only baseline typology (from Iteration 2) is stable at finer $k = 4$ (Jaccard 0.856). After residualising on early volume, the 3-channel typology does NOT separate unclustered outcomes better than the entropy-only baseline (newcomer share ε^2 0.171 vs 0.178, CIs overlap).

4.6.2 Community Roles

Leiden community roles with 5-seed agreement (97.2% robust, placebo 0.87): BRIDGE 0.61, OTHER 0.33, STAYER 0.03, MIGRANT 0.02. Lagged roles do not robustly predict host-subfield entry: BRIDGE OR 0.64 [0.38, 1.08].

4.6.3 Expansion Precedes Diffusion

In 15 of 16 concepts exhibiting both an expansion onset and a diffusion onset, expansion precedes diffusion (proportion 0.94 [0.81, 1.0], year-shuffle null 0.62, $p = 0.004$). This is underpowered (16 concepts) but directionally strong.

4.7 Dead Ends and Negative Results (Iteration 3)

1. **Openness does not predict WHERE concepts travel.** All Holm $p = 0.405$.
2. **Brokerage is not the mechanism behind low closure.** Burt constraint is higher, not lower, before onset ($S = +0.083$ [0.017, 0.147]).
3. **The 3-channel diffusion typology adds nothing beyond entropy.** After residualising on volume, ϵ^2 0.171 vs 0.178.
4. **Community roles do not predict host entry.** BRIDGE OR 0.64 [0.38, 1.08].
5. **Co-transfer is null.** Arriving with origin companions (CT) does not predict establishment in any specification.

5 Iteration 4: Held-Out Confirmation and Cross-Population Replication

5.1 Strategy

Iteration 4 is a CONFIRMATION iteration. Its purpose is to open the four sealed held-out folds and determine which screen-fold findings replicate, to run four discriminating tests (single-paper artefact, vocabulary decomposition, topic-score absorption, and MeSH replication) that separate genuine host-vocabulary effects from artefacts and circularity, to replicate the grafting result on the independent MeSH population, to test whether pre-emergence closure predicts later host-entry anchoring, and to test the adopter-level mechanism. Five artifacts were executed.

5.2 Artifact 16: Held-Out Confirmation of Grafting and Discriminating Tests (gen_art_evaluation_2)

5.2.1 Held-Out Grafting Confirmation

The co-primary specification confirms the grafting effect on the held-out fold: IRR/SD = 1.19 [1.06, 1.33], $p = 0.0045$, wild-cluster $p = 0.012$. The primary specification is inconclusive because it retains only 23% of events (250 of 1,097) with 30 concept clusters. Co-transfer is null on both folds.

Table 20: Held-out grafting results. Source: `gen_art_evaluation_2`, Iteration 4.

Specification	Fold	IRR/SD	95% CI	p	CT	CT p	N	G
Primary	Held-out	0.98	[0.70, 1.37]	0.91	0.82	0.22	250	30
Co-primary	Held-out	1.19	[1.06, 1.33]	0.0045	1.04	0.60	972	74
Co-primary	Screen	1.30	[1.16, 1.45]	1×10^{-5}	–	–	1,544	140

Table 21: Single-paper artefact interaction results. Source: `gen_art_evaluation_2`, Iteration 4.

Fold	IRR/SD (single)	IRR/SD (multi)	Ratio	p (interaction)
Pooled	1.26	1.28	0.92	0.11
Held-out	1.37	1.19	0.82	0.004

5.2.2 Single-Paper Artefact Test

87% of screen co-primary events (1,344 of 1,544) are single-paper entries.

The pooled multi-team effect remains positive (1.28 [1.11, 1.48]), though on the held-out fold, single-paper entries drive the effect more strongly.

5.2.3 Host-Vocabulary Decomposition

Table 22: Vocabulary decomposition (co-primary spec). Source: `gen_art_evaluation_2`, Iteration 4.

Component	Fold	IRR/SD	95% CI	p
NATIVE	Pooled	1.11	[1.05, 1.19]	0.0008
ADJACENT	Pooled	1.32	[1.20, 1.46]	9×10^{-8}
NATIVE	Held-out	1.10	[1.01, 1.20]	0.036
ADJACENT	Held-out	1.19	[1.03, 1.38]	0.020

Both NATIVE and ADJACENT components are positive on both folds. The grafting effect runs through the full host-vocabulary gradient, not just through the native tail.

5.2.4 Classifier Circularity (Topic-Score Absorption)

Combined topic-score absorption: -7.3% $[-27.5, +6.7]$ of the log-IRR on the pooled fold, -10.3% $[-84.3, +56.6]$ on the held-out fold. Topic-score controls do not meaningfully reduce the A_{cont} effect. The mechanism label is **host-vocabulary**.

5.3 Artifact 17: MeSH Replication of Grafting (`gen_art_experiment_9`)

Verdict: REPLICATED. Both specifications are significant on MeSH. The IVW pooled estimate across main-arm co-primary (IRR/SD 1.30) and MeSH co-primary (1.23) is 1.26 [1.17, 1.36] with $I^2 = 0$.

Table 23: MeSH grafting results. Source: `gen_art_experiment_9`, Iteration 4.

Specification	IRR/SD	95% CI	Holm p	CT	CT p	N
R1: primary	1.32	[1.10, 1.60]	0.007	1.10	0.46	1,004
R2: co-primary	1.23	[1.12, 1.36]	9.7×10^{-5}	1.10	0.053	2,171

Table 24: Closure held-out results. Source: `gen_art_evaluation_3`, Iteration 4.

Measure	Screen S	Held-out S	Held-out 95% CI	Holm p	Decision
Closure (pooled panel)	-0.661	-0.391	[-0.855, 0.072]	0.147	DEAD
Turnover-resid. closure	-0.400	-0.109	[-0.561, 0.342]	0.635	DEAD
Persist.-nbr closure	-1.124	-1.073	[-1.858, -0.289]	0.015	CONFIRMED
Burt constraint	+0.083	+0.105	[0.021, 0.187]	-	DEAD

5.4 Artifact 18: Closure Held-Out Confirmation (`gen_art_evaluation_3`)

Overall closure status: not confirmed. The primary closure measure and the turnover-residualised closure both fail. Only persistent-neighbour closure is CONFIRMED on the held-out fold: pooled-panel coefficient -1.073 $[-1.858, -0.289]$, Holm $p = 0.015$ (event-study $S = -0.826$ $[-1.706, -0.096]$). Under the frozen kill mapping RQ1 is R1_DEAD: structural precursors of sustained uptake are volume/churn correlates. Burt constraint replicates its positive sign (+0.105), confirming that the effect is not Burt brokerage.

5.4.1 Closure-Anchoring Link Test

Table 25: Closure-anchoring link test. Source: `gen_art_evaluation_3`, Iteration 4.

Fold	n	Partial ρ	95% CI	p	MDE (ρ)
Screen	102	+0.025	[-0.199, 0.242]	0.806	0.258
Held-out	49	+0.053	[-0.279, 0.391]	0.736	0.380
Pooled	151	+0.049	[-0.126, 0.228]	0.564	0.212

Verdict: NULL. There is no detectable association between pre-emergence closure and later host-entry anchoring. The two findings (closure and grafting) are independent.

5.5 Artifact 19: Descriptive Held-Out Confirmation (`gen_art_evaluation_4`)

5.5.1 Lead-Lag: Expansion Before Diffusion

Among concepts with both transitions observable, expansion precedes diffusion in 25 of 26 pooled main-arm concepts. On MeSH, 9 of 13 dual-onset concepts show expansion first (proportion 0.69 vs year-shuffle null 0.62, $p = 0.43$: not different from the null).

Table 26: Lead-lag category counts. Source: `gen_art_evaluation_4`, Iteration 4.

Category	Screen	Held-out	Pooled main	MeSH
Neither	110 (54%)	49 (49%)	159 (53%)	71 (37%)
Diffusion only	39 (19%)	23 (23%)	62 (21%)	97 (51%)
Expansion only	37 (18%)	18 (18%)	55 (18%)	10 (5%)
Expansion first	15 (7%)	10 (10%)	25 (8%)	9 (5%)
Same year	1 (0.5%)	0	1 (0.3%)	1 (0.5%)
Diffusion first	0	0	0	3 (1.6%)

5.6 Artifact 20: Adopter-Level Mechanism (`gen_art_evaluation_5`)

This artifact tests whether authors who adopt a concept in a new host subfield had prior corpus exposure to the concept’s entry partners, providing an adopter-level mechanism for the host-vocabulary gradient.

Table 27: Adopter enrichment results. Source: `gen_art_evaluation_5`, Iteration 4.

Exposure measure	OR	95% CI (boot)	p (CRV)
E_{any} (any prior partner)	3.09	[2.32, 4.28]	1.4×10^{-13}
E_{neg} (freq.-matched negative)	0.62	[0.49, 0.77]	2.6×10^{-5}
E_{plac} (placebo-host partners)	0.72	[0.57, 0.95]	0.013
E_{swap} (exposure swap)	1.11	[0.86, 1.39]	0.39

Table 28: Vocabulary-class enrichment. Source: `gen_art_evaluation_5`, Iteration 4.

Exposure class	OR	95% CI (boot)	p (CRV)
E_{for} (FOREIGN)	2.88	[2.28, 3.69]	$< 1 \times 10^{-6}$
E_{adj} (ADJACENT)	1.91	[1.43, 2.73]	1×10^{-4}
E_{nat} (NATIVE)	1.47	[1.02, 2.33]	0.059
E_{neg} (non-partner)	0.64	[0.51, 0.82]	0.0003

Adopters have higher odds of prior corpus exposure to the concept’s entry partners than matched non-adopters (OR 3.09 [2.32, 4.28]; exposure prevalence 0.79 vs 0.61, risk ratio about 1.30). The result is consistent with absorptive capacity or with topical proximity. The gradient runs FOREIGN > ADJACENT > NATIVE: adopters are most enriched for exposure to the concept’s foreign (origin-vocabulary) partners. This runs in the opposite direction from the entry-level grafting result and indicates that the two levels operate through different channels.

5.7 Dead Ends and Negative Results (Iteration 4)

1. **Closure not confirmed on held-out for the primary measure.** Raw closure pooled-panel coefficient -0.391 , Holm $p = 0.147$; turnover-residualised closure coefficient -0.109 , Holm $p = 0.635$.
2. **Closure-anchoring link: null.** Partial $\rho = 0.025$ screen, 0.053 held-out.
3. **BROAD concepts do not anchor more.** Screen -0.0018 , held-out $+0.0005$, CI including zero.
4. **Grafting primary specification: inconclusive.** IRR/SD 0.98 $[0.70, 1.37]$ on held-out, 30 clusters.
5. **Single-paper interaction on held-out.** Ratio 0.80 ($p = 0.004$): single-paper entries drive the effect more strongly.
6. **Anchoring $\times$ typology interaction flips between folds.** Not confirmed.
7. **Breadth prediction: all six sensitivity cells null on held-out.**

5.8 Run Ledger (Iteration 4)

Table 29: Run ledger, Iteration 4.

Artifact	Type	Status	Wall time
gen_art_evaluation_2 (grafting held-out)	evaluation	succeeded	~ 7 min
gen_art_experiment_9 (MeSH replication)	experiment	succeeded	~ 12 min
gen_art_evaluation_3 (closure held-out)	evaluation	succeeded	~ 10 min
gen_art_evaluation_4 (descriptive held-out)	evaluation	succeeded	~ 8 min
gen_art_evaluation_5 (adopter mechanism)	evaluation	succeeded	~ 6 min

6 Iteration 5: Post-Confirmation Decomposition and Audit

6.1 Strategy

Iteration 5 is a POST-CONFIRMATION EXPLORATORY iteration. The grafting result (A_{cont} predicts newcomer uptake) was confirmed on the held-out fold in Iteration 4 and replicated on the independent MeSH population. The purpose of this iteration is threefold: (i) run three K-tests that decompose the confirmed finding along dimensions that the pre-registered confirmation did not distinguish (extensive vs intensive margin, host-specific vs generic vocabulary, field boundary); (ii) run a full numerical audit of every number in the report; and (iii) produce a positioning dossier and publication-ready figure specifications.

6.2 Artifact 21: Margin Decomposition and Field-Boundary Test (gen_art_evaluation_6)

6.2.1 Extensive vs Intensive Margin

Table 30: Extensive-margin results (LPM, pp change per SD of A_{cont}). Source: gen_art_evaluation_6, Iteration 5.

Fold	pp/SD	95% CI	p (CRV1)	p (rand- t)	N
Screen	+6.46	[3.42, 9.51]	4.5×10^{-5}	0.0005	1,686
Held-out	+7.29	[2.97, 11.61]	0.0012	0.004	1,046
MeSH	+6.17	[3.87, 8.46]	3.4×10^{-7}	0.0005	2,236
IVW	+6.43	[4.76, 8.11]	5.1×10^{-14}	0.0005	–

A one-SD increase in host-nativeness of entry partners raises the probability that any newcomer uptake occurs by 6.4 percentage points (IVW pooled [4.76, 8.11], $I^2 = 0$).

Table 31: Intensive-margin results (PPML on $Y \geq 1$ subsample, IRR/SD). Source: gen_art_evaluation_6, Iteration 5.

Fold	IRR/SD	95% CI	p (CRV1)	N
Screen	1.272	[1.121, 1.443]	0.0003	795
Held-out	1.163	[1.008, 1.342]	0.038	489
MeSH	1.137	[1.026, 1.260]	0.015	1,169
IVW	1.183	[1.104, 1.267]	1.7×10^{-6}	–

Margin-decomposition verdict: BOTH. Host-leaning entry vocabulary raises both the probability that uptake starts (extensive margin) and its magnitude given it has started (intensive margin). The extensive margin accounts for 38% [21%, 55%] of the total log-IRR.

6.2.2 Field Boundary

Table 32: Field-boundary results (co-primary PPML, IRR/SD by origin field). Source: gen_art_evaluation_6, Iteration 5.

Origin field	IRR/SD	95% CI	p
Physics/Astronomy	1.178	[0.926, 1.499]	0.18
Computer Science	1.216	[1.066, 1.386]	0.004
Other	1.281	[1.167, 1.407]	3.5×10^{-6}
MeSH (reference)	1.233	[1.120, 1.360]	9.7×10^{-5}

Wald test for equality: $W = 0.900$, $p = 0.638$ (CRV1). **No detectable field boundary.**

6.3 Artifact 22: Host-Specificity Test (gen_art_evaluation_7)

Host-specificity verdict: HOST-SPECIFIC. Generality controls do not reduce the A_{cont} coefficient (IVW retention 0.969, excluding 0.5). The Balassa-index A_{lift} replicates across all folds

Table 33: Host-specificity retention: A_{cont} coefficient stability after adding generality controls. Source: gen_art_evaluation_7, Iteration 5.

Fold	Retention	95% CI	% removed	G_H IRR	G_F IRR
Screen	1.006	[0.733, 1.312]	−0.6%	1.122	0.856
Held-out	1.062	[0.438, 1.802]	−6.2%	1.068	0.943
MeSH	0.876	[0.640, 1.037]	+12.4%	1.110	0.764
IVW	0.969	[0.81, 1.10]	+3.1%	1.103	0.815

Table 34: A_{lift} results (PPML, IRR/SD). Source: gen_art_evaluation_7, Iteration 5.

Fold	Model	IRR/SD	95% CI	Holm p
Screen	M2 (A_{lift} alone)	1.668	[1.378, 2.019]	4.4×10^{-7}
Screen	M3 ($A_{\text{lift}} + G$)	1.661	[1.361, 2.026]	1.4×10^{-6}
Held-out	M2	1.457	[1.140, 1.862]	0.003
Held-out	M3	1.459	[1.146, 1.856]	0.003
MeSH	M2	1.286	[1.147, 1.441]	2.5×10^{-5}
MeSH	M3	1.244	[1.120, 1.381]	6.3×10^{-5}
IVW	M2	1.388	[1.268, 1.519]	1.0×10^{-12}
IVW	M3	1.340	[1.232, 1.458]	1.5×10^{-10}

(IVW IRR/SD 1.34 [1.23, 1.46]).

6.4 Artifact 23: Full Numerical Audit (gen_art_evaluation_8)

6.4.1 Decision-Rule Consistency

6.4.2 RQ1 Summary from Audit

6.4.3 Caveats Summary from Audit

6.4.4 Adopter Enrichment from Audit

6.4.5 MeSH Replication Caveats

6.4.6 Rooting Contrasts

6.4.7 Representative Cases

Four medoid cases were selected from the typology clusters.

Case 1: Wireless backhaul (BROAD medoid, Computer Science, $F = 2006$). Wireless backhaul is broad from its first year. It has 14 host entries, all in the co-primary sample. Only one entry is rooted (Aerospace Engineering in 2008, $A_{\text{cont}} = 0.084$, graft label anchored, 6 newcomer papers). The 13 non-rooted entries average $A_{\text{cont}} = 0.021$, and most arrive as origin packages.

Case 2: Einstein-Podolsky-Rosen steering (BROAD, nearest non-medoid, Physics, $F = 2011$). Of its 7 co-primary entries, one is rooted: AI in 2011 ($A_{\text{cont}} = 0.142$, anchored, 37 newcomer papers). Rooted-minus-unrooted A_{cont} is +0.115.

Case 3: Locally repairable code (LOCALISED medoid, Computer Science, $F = 2013$). Stays at 84–96% in Computer Networks and Communications. Only one co-primary entry (AI in 2013, $A_{\text{cont}} = 0.056$), which is rooted.

Table 35: Audit module results. Source: gen_art_evaluation_8, Iteration 5.

Module	Description	Result
M1	Curated-number accuracy	99.4% (319 curated)
M2	Headline + independent check	90.8% (210 R-tier 92.9%, 50 C-tier 82.0%)
M3	Rederivation by block	D2: 80 OK, 1 WRONG_ESTIMATOR, 16 MISSING. MeSH: 29 OK, 5 MISSING. RQ1: 60 OK, 1 WRONG_ESTIMATOR, 10 MISSING
M4	Drift detection	45 drift rows across 32 lines; 6 VERDICT-CHANGING
M5	Seeded-perturbation recall	Overall 0.60; CI swaps 1.0, verdict flips 0.9
M6	Independent claim review	73/73 agree; 0 blocked claims
M7	Table-cell traceability	11/11 tables pass
M8	Reviewer-critique closure	All 6 MAJOR items CLOSED
M9	Decision-rule consistency	13 rules; 1 mismatch
M10	Flag inventory	292 rows flagged; 8 SENS2 flags
M11	Self-test honesty	4 PARTIAL, 4 NOT DONE, 3 DONE

Table 36: Reviewer-critique closure verified by audit. Source: gen_art_evaluation_8, Iteration 5.

#	Critique	Status	Closed by
1	RQ1 headline contradicts frozen kill rule	CLOSED	T_rq1, T_design
2	Table 18 claims fixes not made	CLOSED	T_table18, drift.csv
3	Grafting caveats dropped	CLOSED	T_caveats, drift.csv
4	MeSH and adopter read too broadly	CLOSED	T_mesh, T_adopter
5	RQ2 descriptive results missing	CLOSED	cases.md, T_rooting
6	Decision rules not recorded	CLOSED	T_decision

Table 37: Decision-rule consistency (selected rows). Source: gen_art_evaluation_8, Iteration 5.

Iter	Rule	Outcome	Status
2	(a) Gate A $\rightarrow$ graft fallback	TRIGGERED (0.17 < 0.50)	MATCH
2	(b) H1 pilot-only if < 150 concepts	YES ($N_c = 38$)	MATCH
2	(c) RQ1 CONFIRMED conditions	NOT MET	MATCH
3	R1 mechanical verdict	MIXED	MATCH
3	D1 SUPPORTED on screen	NOT SUPPORTED	MATCH
4	D2 DEAD if held-out CI includes 1	NOT DEAD	MATCH
4	Mechanism label	host-vocabulary	MATCH
4	G4 second-family replication	REPLICATED	MATCH
4	R1 dead if held-out row fails	R1_DEAD (Holm 0.147)	CORRECTED

Table 38: RQ1 results summary (selected rows from T_rq1). Source: gen_art_evaluation_8, Iteration 5.

Test	Value	Source
Closure (R1_DEAD)	Holm $p = 0.147$	r1_holm_decisions.csv
Closure_persist	CONFIRMED (Holm $p = 0.015$)	r1_holm_decisions.csv
Constraint	screen 0.083, held-out 0.105	r1_event_study
RQ1 VERDICT	R1_DEAD	—

Table 39: Inferential caveats (selected rows from T_caveats). Source: gen_art_evaluation_8, Iteration 5.

Test	Value	Source
CRV1 p (co-primary A_{cont})	0.0045	heldout_post.json
Wild-cluster bootstrap p	0.012	heldout_post.json
Nativeness-permutation p (500 draws)	0.026	heldout_post.json
Within-concept A -shuffle perm p	0.050	audit_perm.json
CRV1 rejection rate under null	17.5%	audit_perm.json
EST_bin LPM A_{cont} p (co-primary FE)	0.165	heldout_post.json
Primary FE IRR/SD ($G = 30$, inconclusive)	0.98	heldout_post.json
Screen co-primary single-paper share	87.0%	recount

Table 40: Adopter enrichment (selected rows from T_adopter). Source: gen_art_evaluation_8, Iteration 5.

Exposure	OR	95% CI	Note
E_{any} (primary m2)	3.09	[2.32, 4.28]	prevalence 0.79 vs 0.61
E_{neg} (freq.-matched neg.)	0.62	[0.49, 0.77]	
E_{plac} (placebo-host)	0.72	[0.57, 0.90]	
E_{for} (FOREIGN)	2.88	[2.28, 3.69]	
E_{adj} (ADJACENT)	1.91	[1.43, 2.73]	
E_{nat} (NATIVE)	1.47	[1.02, 2.33]	
$E_{\text{any}} \times A_{\text{cont}}$ interaction	0.87	[0.70, 1.14]	

Table 41: MeSH replication caveats (selected rows from T_mesh). Source: gen_art_evaluation_8, Iteration 5.

Item	Value	N
Coverage rule removes share of entries	34.1%	–
R1 primary FE IRR/SD	1.324	1,004
R2 co-primary IRR/SD	1.233	2,171
S3 placebo host IRR/SD	1.031	2,171
Exact-profile nativeness only	1.243	2,171
S6 coverage-0.50 subset (declared)	1.007	123
G1 multi-team IRR/SD	1.079	830

Table 42: Rooting contrasts by diffusion type. Source: gen_art_evaluation_8, Iteration 5.

Type	Entries/concept	EST rate	Raw A_{cont}	CT	PPML IRR/SD	95% CI
BROAD	18.4	0.31	0.039	0.584	1.36	[1.20, 1.55]
LOCALISED	6.6	0.13	0.056	0.691	1.16	[0.98, 1.38]

Table 43: Per-case entry statistics (co-primary sample). Source: gen_art_evaluation_8, Iteration 5.

Concept	Entries	Rooted	Mean A_{cont} (rooted)	Mean A_{cont} (unrooted)	Diff
Wireless backhaul	14	1	0.084	0.021	+0.062
EPR steering	7	1	0.142	0.027	+0.115
Locally repairable code	1	1	0.056	–	–
Holographic QCD	5	0	–	0.041	–

Case 4: Holographic QCD (LOCALISED, nearest non-medoid, Physics, $F = 2006$). Stays at 95–97% in Nuclear and High Energy Physics. Its 5 co-primary entries are all non-rooted, with three having $CT = 1.0$: pure origin packages that do not take root.

6.5 Artifact 24: Positioning Dossier (gen_art_research_2)

The novelty verdict is PARTLY ANTICIPATED (0 ANTICIPATES, 11 PARTLY among 25 graded candidates). No located work measures how host-specific the vocabulary is that a concept is combined with when it enters a new subfield, and relates that, within concepts and across hosts, to later uptake by the host’s own newcomers. The paper-ready novelty sentence: “To our knowledge, no prior study tests, within concepts and across the host subfields they enter, whether the host-specificity of the vocabulary a concept is first combined with predicts uptake by the host’s own newcomers.”

6.6 Artifact 25: Paper Figures (gen_art_evaluation_9)

Table 44: Figure inventory and production checks. Source: gen_art_evaluation_9, Iteration 5.

Figure	Description	Values checked	All pass
F1	Method flow (study design overview)	39 exact	Yes
F2	D2 forest plot (grafting across folds)	123 exact	Yes
F3	Mechanism panel (vocab. decomp. + adopter)	162 exact	Yes
F4	RQ1 held-out (closure, R1_DEAD stated)	94 exact	Yes
F5	Occupancy-rooting (typology $\times$ anchoring)	67 exact	Yes
F6	Case studies (4 medoid concepts)	425 exact	Yes
F7	Self-test (audit pass/fail summary)	0 (schematic)	Yes

All 8 figures pass production checks: correct dimensions (174 mm width), 600 DPI, no Type 3 fonts, colourblind-safe palettes. Total: 957 values checked, 0 failures, value fidelity 1.0.

6.7 Dead Ends and Negative Results (Iteration 5)

1. **EST_bin null on held-out.** The binary establishment threshold (≥ 3 newcomer papers and host papers in ≥ 3 of 5 years) shows +2.54 pp per SD on the held-out fold with $p = 0.16$. The claim holds for uptake initiation ($Y \geq 1$) but not for strict establishment.
2. **Intensive margin is conditional, not causal.** The margin-decomposition intensive estimate conditions on $Y \geq 1$, subject to selection bias.
3. **Field boundary not detected.** Wald $p = 0.638$.
4. **MeSH retention slightly lower.** On MeSH, 12.4% of the A_{cont} effect is absorbed by field breadth.
5. **Audit reveals 292 MISSING IN REPORT rows.** Numbers present in artifact output files but not transcribed into the report.
6. **Seeded-perturbation recall 0.60.** Estimator/fold relabelling detection at 0.10.

Table 45: Run ledger, Iteration 5.

Artifact	Type	Status	Wall time
gen_art_evaluation_6 (K1/K3 tests)	evaluation	succeeded	~12 min
gen_art_evaluation_7 (K2 test)	evaluation	succeeded	~10 min
gen_art_evaluation_8 (full audit)	evaluation	succeeded	~70 s
gen_art_research_2 (positioning dossier)	research	succeeded	–
gen_art_evaluation_9 (paper figures)	evaluation	succeeded	–

6.8 Run Ledger (Iteration 5)

7 Run Ledger (All Iterations)

Table 46: Complete run ledger across all iterations.

Artifact	Iter	Type	Status	Credits
gen_art_dataset_1 (pool)	1	dataset	succeeded	2,566
gen_art_dataset_2 (background)	1	dataset	failed (timeout)	1,952
gen_art_dataset_3 (MeSH)	1	dataset	succeeded	1,784
gen_art_dataset_4 (grounding)	1	dataset	succeeded	1,847
gen_art_research_1 (dossier)	1	research	succeeded	–
gen_art_dataset_5 (hydrated)	2	dataset	succeeded	7,237
gen_art_experiment_1 (grounding)	2	experiment	succeeded	–
gen_art_experiment_2 (viability)	2	experiment	succeeded	–
gen_art_experiment_3 (emergence, main)	2	experiment	succeeded	–
gen_art_experiment_4 (emergence, MeSH)	2	experiment	succeeded	–
gen_art_evaluation_1 (audit)	3	evaluation	succeeded	–
gen_art_experiment_5 (emergence deepen)	3	experiment	succeeded	–
gen_art_experiment_6 (breadth)	3	experiment	succeeded	–

Artifact	Iter	Type	Status	Credits
gen_art_experiment_7 (grafting)	3	experiment	succeeded	–
gen_art_experiment_8 (diffusion)	3	experiment	succeeded	–
gen_art_evaluation_2 (grafting held-out)	4	evaluation	succeeded	–
gen_art_experiment_9 (MeSH replication)	4	experiment	succeeded	–
gen_art_evaluation_3 (closure held-out)	4	evaluation	succeeded	–
gen_art_evaluation_4 (descriptive held-out)	4	evaluation	succeeded	–
gen_art_evaluation_5 (adopter)	4	evaluation	succeeded	–
gen_art_evaluation_6 (K1/K3)	5	evaluation	succeeded	–
gen_art_evaluation_7 (K2)	5	evaluation	succeeded	–
gen_art_evaluation_8 (full audit)	5	evaluation	succeeded	–
gen_art_research_2 (positioning)	5	research	succeeded	–
gen_art_evaluation_9 (figures)	5	evaluation	succeeded	–

8 Pre-Registration Hashes

9 What the Run Learned Overall

9.1 RQ2: Diffusion — The Grafting Effect Is Confirmed and Replicated

The headline finding is that when a scientific concept enters a new disciplinary subfield, the host-vocabulary share of its initial co-occurrence partners predicts subsequent uptake by newcomer authors. This was confirmed on the independent held-out fold (co-primary IRR/SD 1.19 [1.06, 1.33], $p = 0.004$) and replicated on the independent MeSH biomedical population for biomedicine-to-biomedicine host entries (co-primary IRR/SD 1.23 [1.12, 1.36], Holm $p < 0.001$). The IVW pool of the main-arm screen and MeSH co-primary estimates is 1.26 [1.17, 1.36] with $I^2 = 0$. Co-transfer of origin companions is null across all folds and populations.

Post-confirmation decomposition shows:

- **Both margins matter.** The effect operates through both the extensive margin (IVW +6.43 pp/SD [4.76, 8.11]) and the intensive margin (IVW IRR/SD 1.183 [1.104, 1.267]).
- **Host-specific, not generic.** Generality controls leave A_{cont} unchanged (IVW retention 0.969 [0.81, 1.10]). The Balassa-index A_{lift} replicates across all folds (IVW IRR/SD 1.34 [1.23, 1.46]).
- **No field boundary.** Wald $p = 0.638$.

Table 47: Pre-registration and specification hashes.

Item	SHA-256 prefix
Frozen concept frame	80e3f244...0c44
Viability pre-registration	5dd16d8a...5aac
Test population	6a887fb4...5505
R1a turnover pre-check spec	60f44c0f...
Emergence deepen spec (SPEC3)	2f46d173...
Host-entry grafting pre-registration	f800a0a9...
Descriptive diffusion held-out spec	695166b4...
Held-out confirmation spec (exp_5)	535c2dd3...
Grafting held-out spec	8db17113...
Grafting held-out confirmation spec	4100e3cf...
Discriminating-test spec	a10b7f31...
MeSH replication spec	b7cdabf8...
Closure-anchoring link spec	5d0144d7...
Descriptive held-out spec	4ce330d3...
Adopter mechanism spec	96e697c6...
K1/K3 spec	2435f909...
K2 spec	866c60a8...

- **Adopter mechanism.** Adopters have higher odds than matched non-adopters of prior exposure to the concept’s entry partners (OR 3.09 [2.32, 4.28]; prevalence 0.79 vs 0.61), with enrichment strongest for FOREIGN partners (OR 2.88); consistent with absorptive capacity or topical proximity (screen only).

Caveats: the primary fully-saturated specification is underpowered on the held-out fold (IR-R/SD 0.98, 30 clusters). Binary establishment (EST_bin) is null on the held-out fold ($p = 0.165$). Single-paper entries (87% of screen events) drive the effect more strongly on the held-out fold. The Physics/Astronomy stratum is individually non-significant, but it is within sampling variation of the other groups.

9.2 RQ1: Emergence — Primary Closure Dead, Persistent-Neighbour Confirmed

The general closure deficit did not survive held-out confirmation (R1_DEAD: Holm $p = 0.147$): structural precursors of sustained uptake are volume/churn correlates. Only persistent-neighbour closure is confirmed (held-out pooled-panel coefficient -1.07 , Holm $p = 0.015$). This is a narrower claim: emerging concepts have lower triadic closure among their stable top associates, but the general closure deficit is not robust to the held-out test. The effect is not Burt brokerage (constraint is higher, not lower). Pre-emergence closure does not predict later host-entry anchoring (closure-anchoring link null), so the two findings are independent.

9.3 Diffusion Typology

A data-derived $k = 2$ diffusion typology separates localised ($n = 136$) from broad-from-the-start ($n = 66$) concepts. Finer typologies are unstable. Network expansion precedes disciplinary diffusion in 25 of 26 pooled main-arm dual-onset concepts. BROAD concepts have lower co-transfer but do not anchor more.

9.4 What Is Still Open

1. **Causal identification.** The grafting result is associational. Concepts that enter with host-native partners may differ from concepts that enter with foreign partners on unobserved dimensions.
2. **Generalisation beyond physics/CS/biomedicine.** The main-arm pool is dominated by arXiv-sourced concepts. Social sciences and humanities are untested.
3. **Binary establishment.** The strict establishment threshold (≥ 3 newcomer papers and host papers in ≥ 3 of 5 years) is null on the held-out fold. The claim holds for uptake initiation but not for sustained establishment.
4. **Single-paper artefact.** 87% of co-primary events are single-paper entries. The held-out interaction shows single-paper entries driving the effect more strongly.
5. **The intensive margin is conditional.** The intensive-margin estimate conditions on $Y \geq 1$ and is subject to selection bias.
6. **Adopter-level mechanism ambiguity.** The design cannot separate absorptive capacity from topical proximity.
7. **292 MISSING IN REPORT rows.** Numbers present in artifact output files but not transcribed, spanning second-order sensitivity analyses.